

UnionLabs: AWS-based Remote Access and Sharing of NextG and IoT Testbeds

UnionLabs is a cloud-based platform that makes wireless research testbeds remotely accessible and shareable, so that experiments on NextG and IoT hardware can be run from a browser by users at any institution. The core mechanism is already prototyped: a reservation is translated automatically into an isolated environment with the reserved radios attached, and is served to the user as a remote desktop. Three components remain to be built on top of it: a web front end providing user accounts, a catalog of available resources, a reservation calendar, and session management; hardware integration that brings additional device classes, in particular IoT nodes, under the same reservation model; and a library of ready-to-run wireless experiments that a remote user can launch to verify a reservation. Most of the platform is already in place, so a senior project team would build on existing work rather than start from scratch. The students would first study the existing system and codebase, covering Python, JavaScript, and C++, together with the basic wireless background. They would then take the web design as an independent mission, with the implementation of wireless experiments as an optional additional task.